



Examiners' Report

Summer 2026

Pearson Edexcel GCE

In AS Further Mathematics (8FM0)

Paper 28 Decision Mathematics 2

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General Comments

This paper proved demanding but accessible to many learners. Many demonstrated sound knowledge across much of the content and produced well-presented solutions using appropriate techniques, tables and diagrams. Some responses were more difficult to follow, so learners should be reminded that Decision Mathematics is a methods-based examination in which clear presentation of method is essential.

Questions 4 and 5 gave learners opportunities to demonstrate higher-level reasoning and problem-solving skills. While many learners made progress on the recurrence relations question, this remained one of the more demanding topics on the paper. The final question also proved challenging for many learners.

Comments on Individual Questions

Question 1

This question was well answered, with many learners able to provide fully accurate responses (including the use of an appropriate and well-drawn diagram).

The first B mark was scored for the original setup of the problem; however, many candidates did not define p and $(1 - p)$ correctly. Most responses showed knowledge of the required techniques; however, learners should be reminded to fully define any variable they introduce into a problem. Beyond this, most learners were able to form the correct algebraic expressions, draw the correct graph and solve for the correct point, then give a conclusion in context to the problem. Some learners drew the correct graph but selected the incorrect point, and a very small proportion of responses selected the wrong two expressions to equate. However, for the most part this was a very accessible opening question which scored well for most learners.

Question 2

A good proportion of responses showed understanding of the required process and a sound grasp of the mechanics of the Hungarian Algorithm. Some learners introduced a dummy column using a large value in each entry rather than zeroes. This approach often led to errors in the subsequent stages of the algorithm, so learners should be reminded that dummy entries should be handled carefully and consistently.

Most learners were able to reduce columns (and rows if required) correctly, though some learners did not access the marks by attempting to reduce rows ignoring the zeroed dummy column. Nearly all learners who reduced rows and columns correctly were able to demonstrate the correct method for augmenting their table at each step, accepting that some numerical slips were made by some learners. Nearly all learners who scored the three method marks were also able to give a correct allocation and state the minimum time correctly. A very small proportion of learners attempted to convert this into a maximisation problem incorrectly; these attempts generally scored almost no marks as they were followed by further errors in attempting to demonstrate the algorithm.

Question 3

Nearly all learners scored the mark in (a) for completing the initialisation of the flow-augmentation procedure in their answer book.

Responses to part (b) were mixed. Some learners chose routes that could not be used to augment the flow, while others did not apply the labelling procedure correctly. Learners should be reminded to check whether arcs are already at maximum capacity, as no additional flow can be sent along these arcs. Only a minority of learners were able to successfully augment the flow by 9 units. Learners who successfully augmented the flow generally showed the improved flow on the diagram. Some responses included a further labelling procedure diagram instead. Learners should be reminded that, when asked to show the flow through a network, they should add only a single value to each arc. Including both the capacity and the flow, for example by circling the flow, can make the answer ambiguous.

It was clear that many learners who could not find the correct flow augmentation were aware of the Maximum Flow – Minimum Cut theorem but did not demonstrate an improved flow and correct cut to score these marks. Among the responses that successfully augmented the flow by 9 units, many were also able to find a correct cut and state the required information. To complete this part fully, learners needed to state the value of the increased flow, the value of the cut found, and then give their conclusion. Some responses gained only the method mark because either the value of the cut or the value of the flow was not stated.

Question 4

This question proved challenging for many learners. A reasonable proportion of responses went no further than stating the complementary function. Some responses recalled the correct form for the particular solution, but the substitution of this expression, using n and $n + 1$, into the recurrence relation to find the value of λ was less often completed successfully.

Responses that found a correct value for λ were often able to complete the full solution. In part (b), some responses considered only the first few terms by substitution and incorrectly concluded that the sequence was positive for all

values. This approach did not provide a sufficient general argument and therefore did not gain credit.

Question 5

Parts (a) and (b) were reasonably well attempted by many learners, although there were some non-attempts in part (a). Many responses used the information provided by the completed Row Minimum column in the answer book to give the required column maximums in the correct form. A good proportion of responses also stated the correct condition for a zero-sum game to have a stable solution. In part (b), some responses gave ambiguous statements such as maximin = minimax, which is not sufficient to score this mark.

Part (c) proved to be highly challenging. Many responses did not attempt this part, and some of those that did set up equations or inequalities without fully addressing the different possible cases. Very few responses identified that the value of the game could be either 3 or $5 - 2k$, with most focusing only on $5 - 2k$. Some responses considered the conditions for $5 - 2k$ to be the stable solution, but without considering the alternative case of 3, a complete solution could not be achieved even when a correct range for k was given.

Summary

- Learners should be reminded of the importance of good basics, such as defining p and $1 - p$ clearly in a game theory question.
- Learners should continue to develop their understanding of recurrence relations.
- Learners should ensure that they are familiar with the requirements of network flow problems, including the use of the labelling procedure in flow augmentation.
- Learners should be encouraged to give worded explanations with sufficient detail and clarity, avoiding ambiguous statements such as those noted in Question 5.